

1 Dynamic Dialogs - Fields

Overview

Menu	System administration → Terminals → Dynamic dialogs, Tab <i>Dynamic Dialogs - fields</i>
Transaction code	ddconf
Function authorization	ddconf

Other option:

Menu	System administration → Terminals → Dynamic dialogs - fields
Transaction code	ddconf
Function authorization	ddconf

Purpose

You can use the configuration of the dynamic dialogs to change the AIP input dialogs in a quick and efficient manner according to the user's requirements.

The system delivery includes a basic dialog configuration. You can edit and change this configuration using the functions described in the following.

You can edit the dialog fields in two places: in the tab integrated in the application *Dynamic dialogs* and in the application *Dynamic dialogs - fields*. You can change existing fields (e.g. positioning), create new fields or delete fields.



The complete functionality of the dialog configuration includes a lot of options, but it is also very complex. For this reason, we recommend to change the dialogs only after consultation with MPDV and only by experts.

Integration

To define the tab order, the dialogs and the buttons, you need not only use the application *Dynamic dialogs - fields*, but also the applications *Dynamic dialogs - Workflow*, *Dynamic dialogs* and *Dynamic dialogs - Function keys*.

Requirements

The dialog must exist in the *Dynamic dialogs* function.



Some of the functions require the development license MDS-AIS to be fully available. Basics of the functions without development license:

- Existing data can be changed. Only data for default dialogs with user 0 must not be changed without development license.
- You cannot create new data without development license, except fields in existing dialogs.
- You can copy the existing data to terminal groups or terminals and then change them. Without development license, you cannot copy to default dialogs with user 0.

With the fields of the dynamic dialogs, all functions are available without development license. But without development license, you cannot create, edit or delete default dialogs for user 0.

Selection criteria

The application provides the following selection criteria. The selection criteria are read-only. The values entered in the previous application are automatically preassigned.

Dialog

Selection by dialog

Type

Selection by dialog types:

- AIPDEF/DEF – standard dialog
- AIPTNR/TNR – terminal dialog
- AIPTGRP/TGRP – dialog for terminal group

User

Selection by terminal number or terminal group

Editing functions

The application "Dynamic dialogs - fields" only provides the usual editing functions: insert, edit and delete.

If you use the application *Dynamic dialogs*, you can change to the toolbar tab *Dynamic dialogs - fields*. This toolbar provides the usual editing functions and additionally the detail application *Edit fields*. Using this editing application, you can easily manage and edit the fields of a dialog.

Detail application *Edit fields*

In the detail application *Edit fields*, you can right-click the table view to open a **context menu**. The context menu provides the following functions:

New row

Inserts a new row in the grid for the definition of a field.

Several new rows

Inserts the specified number of rows in the grid.

Copy row

Copies the currently select row and inserts it in the grid.

Delete row(s)

Deletes the currently selected row(s) from the grid.

Swap fields

The selected rows are swapped. You can select one of the two methods:

- **Swap positions**
Swaps the X and Y positions of texts, fields and units of the two selected entries.
- **Swap field numbers**
The field numbers and the tab order of the fields are swapped.

Align fields

Automatically aligns function keys in the X or Y direction.

Move fields

Moves one or several fields in the X or Y direction. Buttons are moved using the specified offset ("Move by").

Apply fields from other dialog

Takes over several fields from the selected dialog.

Field description

Tab General**Activated**

This option is only available for reasons of downward compatibility. Always enable the option.

Field no.

Consecutive number of input field in dialog. Specifies the tab order in the dialog.

Text

Text label of field (on the left of the field).

Unit

Text for unit (on the right of the field).

Information

Information text displayed in tooltip if focused with mouse (mouseover).

Identifier

Identification of the data field in the dialog data string, e.g. ANR → |ANR=123456780100|

ID index

Index of field identification for similar data fields, e.g. |EGR:GUT=12340|.

Tab Position

X pos. text

X position of field label

Y pos. text

Y position of field label

X pos. field

X position of data field

Y pos. field

Y position of data field

Unit X pos.

X position of text for unit

Unit Y pos.

Y position of text for unit

Tab Format

Alignment

Alignment of text in input field.

"L" left left-aligned

"R" right right-aligned

Category

INPUT Field is an input field.

TEXT Alphanumeric text

GRID Table

OPTION Option field

RADIO Selection group (as of CTAIP: see "field attribute 1-8" → "EXTENDED")

Input type

Input type of data field.

ALPHA	alphanumeric
NUMERISCH	numeric without decimal places
FLIESS	numeric with decimal places
DATUM	date
ZEIT	time (00:00:00 - 23:59:59)
DAUER	time period (00:00:00 - 9...9:59:59)

Alternatively, a you can edit a preconfigured field type for user fields in the input type using the selection list, for example ANR. Only MPDV can change the field type for user fields.

Length

Total length of input field in characters.

Formatting

If a field type for user fields is entered in the "Input type" field, the formatting is done using the formatting rules defined for the field type.

If you use the simple input types (FLIESS,...), you can specify in field *Formatting* how the contents are displayed. Exceptions: types NUMERISCH and ALPHA; you cannot store a format for these types.

Sample configurations of input fields of the different input types:

FLIESS	###,### or -###,### or #####	with or without algebraic sign. The decimal separator must be "," or ".".
DAUER	hhhh:mm:ss	maximum 99999:59:59, h: hours with leading zeros
	-dddd:mm:ss	maximum 9999:59:59 + sign,
	d:	hours without leading zeros,
	-:	sign allowed.
	dd,iiii:	industrial format
DATE	dd.mm.yyyy	Default for date fields (need not be specified)
TIME	hh:mm:ss	Default for time indications
	hh:mm	Time without seconds (with leading zeros)
		(must be specified (mandatory))

Allowed characters

You can restrict the characters that can be used. The restriction only applies for the category "INPUT".
If no characters are entered, there is no restriction.

e.g. A-Za-z0-9

Default

You can store a default value that is preassigned to the input field.

e.g. 1.5.

From

Lower limit of default (only relevant for EINGABE category)

To

Upper limit of default (only relevant for EINGABE category)

Optional field 1

Only relevant for OPTION category

Optional field 1: Value, if active

Optional field 2

Only relevant for OPTION category

Optional field 2: Value, if inactive

Radio button

Labels and values for radio buttons (only relevant for RADIO category),

For example: J:Yes:F7;N:No:F8;V:Maybe:F9

1st Value: Return value (identification = X), if selected

2nd Value: Text next to radio button

3rd Value: Function key (optional)

On the AIP, the function keys are automatically assigned to the dialog buttons using the sequence displayed. If the function key of the radio button is additionally configured as function key of a dialog button that triggers actions or is automatically assigned on the AIP, then the dialog button takes priority over the radio button.

Tab Functions**Field attribute 1 to 8**

Field attributes for input fields

- You can use "@AKRONYM/KENNUNG@" to configure an alternative identification to initialize dialog variables.

Application example: (dialog "TRANRLIST")

Dialog field/ID "MATPUF". The column "MPUFF" of the MNR.LST file includes the value initializing the field.

The field will be initialized properly when opening the dialog, if the field "MATPUF" is configured with field attribute "@MNR.MPUFF@". An additional terminal script is not required.

- If the field attribute "DATA.LEN=xxx" is configured, the input length of a field can be changed. The display length is not affected. Example: DATA.LEN=40 (input length 40).
- MANUELL – Input field is visible and editable
- STATUS – Input field is visible but not editable
- NULL - Input field may remain empty and is assigned with default value
- FOCUS - Input field is focused when the dialog is shown
- FOCUSNEXTFIELDONBARC – After editing the field using a serial barcode, the subsequent field is focused.
- BARCODE - Data can only be entered via barcode (keyboard locked)
- READONLY - This field cannot be edited.
- SETVALUE - if configured, the value included in the "default" field is used as the default value, e.g. for statuses (ID "MST") in the "log on OP" dialog. Without this identification, the default value is not written into the field. Note the following on the processing:
If a field is configured several times with SETVALUE (i.e. the same field is included in more than one tab of a workflow), the configuration with the lowest tab index "takes priority".
If a field is configured several times with SETVALUE on one workflow/tab page, the configuration with the greatest field index (no.) takes priority.
- UPPERCASE – Data input is converted into capital letters.
- PASSWORD – Input without visible characters ('***') - transmission is not encrypted
- PWD - Input without visible characters ('***') - transmission is performed with Blowfish encryption
- PWDRSA - Input without visible characters ('***') - transmission is performed with RSA encryption (only AIP2 as of V# 8.2.1.10 and hypdm32.dll V# 8.2.1.24)
- EMPTY – For NUMERIC fields only: if the field value is deleted, an empty input field is displayed instead of value 0.
- UNSELECT – prevents the whole field content from being selected, if the field is focused.
- DIALOGLISTE - modal list dialog that can be called (button behind input field)
Dialog list function must be filled (see below)
- COMBOBOX – Field contains combobox; Combobox function must be filled. (Is not used on the terminal).

General field attributes:

- AUTO - Input field is not visible on the terminal, but field ID is assigned
- LABELFONT – Font type and background color of label are used to display the input field. In combination with the NOBORDER parameter, an input field can be created that is displayed as a label.
- NOBORDER - The input field is displayed without depth effect.
- FIELDLABELFONT – The input field font and background color do not depend on the label. Font type and background color are displayed as configured in the file "dialog.ini" (section "layout", values of "FieldLabelFont") (as of CTAIP).

- **COLORLABELFONT** – The input field font and background color do not depend on the label. Font type and background color are displayed as configured in the file "dialog.ini" (section "layout", values of "ColorLabelFont") (as of CTAIP). For further information, refer to the section "Further descriptions", paragraph "[Field attribute <COLORLABELFONT>](#)".
- **FIELD** – shows variable "data" labels. Select the category "TEXT". The "identification" field must be filled (only CTWIN).
- **EXTENDED** – If configured in the field attribute, you can configure further properties with category <RADIO> in the following fields as of AIP:
 - Number of columns in field "length"
 - Width in field "X position of unit"
 - Height in field "Y position of unit"
- **AUTOTAB** – If the field is completely filled with characters, the cursor goes to the next input field. Example: If a barcode reader is connected via keyboard, the scan can take up several input fields (only CTWIN).

Field attributes of category GRID:

- **<XXX>_GRID** – Definition/function of table (<XXX> = variable; setting only by MPDV)

Entry	Description
AG_GRID	Table with running operations at the machine selected (A_AUT_HU, A_AUT_MPL, A_AUT_RF).
C_MG_BLZ_GRID	Table with quantity info on the input batches for the operation of the machine selected (C_MG_BLZ).
CA_INFO_GRID	Table with batches preceding the operation selected (C_VLOS_MPL, C_VLOS_RF, C_VLOS_S)
CE_ASW_GRID	Table with components of the operation selected (CE_ASW_RF).
CE_GRID	Table of components/input batches of the operation selected with processing (A_AN_[MPL,RF,S], A_P_AN_[MPL,RF], CE_WL[MPL,RF,S]).
CE_INFO_GRID	Table of components/input batches of the operation selected for display (CA_WL_MPL).
FHM_GRID	Table of the resources used at the selected machine (RES_WL).
KOMP_VERB_GRID	Table with components of the selected operation to enter discrete consumption (A_VERB).
PAL_GRID	Table to display the batches of a pallet (C_PALETTE).
SCRIPT_GRID	Table for the display of variable contents. The configuration/processing is defined in the relevant dialog script.
WF_GRID	Table for the display of variable contents. Here, the configuration/processing is performed via the configuration in the section (field <i>Dialog list function</i>) in the layout file (ctwinlay.ini/ctaiplay.ini).

Dialog list function

Function for the dialog list, which can be called on the terminal for dynamic dialogs.

Available dialog lists (entry in the LISTE field; case sensitive):

MNR_LISTE:	List of operations running at the machine
VAG_LISTE:	Order sequencing list (only CTWIN)
VMST_LISTE:	Machine status list
VGGRD_LISTE:	List of deviation reasons
VAGRD_LISTE:	List of scrap quantity reasons
VNCH_LISTE:	List of rework reasons (as of ADE 7.2/ MW2.0)
VPRB_LISTE:	List of problem quantity reasons (as of ADE 7.2/ MW2.0)
VLPKZ_LISTE:	List of premium indicators
VBPOS_LISTE:	List of operator positions
ZLO_LISTE:	List of material buffers (MPL)
TPE_LISTE:	List of transport units (MPL)
HZTYP_LISTE:	List of material types (MPL)
REQRES_LIST :	List of the allowed and assigned resources of a required resource (only AIP2)

Combobox function

Leave this field empty with current configurations. This option is only available for backward compatibility reasons and was used for selection lists on the clients from older system versions (before MW 3).

Select file

File used to read the list (only relevant in combination with the *Combobox function*).

Function 1

Function started upon entering the field, called by parameter list in/out.

Function 2

Function started upon leaving the field, called by parameter list in/out

Options

Blocked

If the field is blocked, it won't be made available by the AIP when the dialog is activated. Therefore, it is unknown in the AIP.

Visible

Invisible fields are processed in the AIP but not displayed. Invisible fields can be used to send fixed acronyms from the AIP to the server. A target value should be added to invisible fields and the field attribute SETVALUE.

Visible/invisible with dialog control

The meaning of this field/control changes whether the option *Visible* is set or not.

- 'Visible' set > Not visible when dialog control is enabled
- 'Visible' not set > Visible when dialog control is enabled

Depending on how the "*Visible*" option is set, you can use identifiers to specify when the field/control is displayed or not.

For further information, refer to the section "Further descriptions", paragraph "[Dialog control](#)".

DB table 1

DB tables, reserved for customization.

DB field 1

DB fields, reserved for customization.

DB table 2

DB tables, reserved for customization.

DB field 2

DB fields, reserved for customization.

User defined 1

Additional configuration options:

If your customer documentation includes no other specification, enter AUTO in this field.

User defined 3

Additional configuration options:

If you configure "FLD.LEN=xxx", you can configure a display length of a field that deviates from the standard, e.g. FLD.LEN=17 (display/input length 17 digits).



Customers often need this configuration to enter longer, customer-specific batch numbers. The input length of the default batch number is configured in the "length of batch no." field in the basic parameter settings.

Further descriptions

Dialog control

Configurations

A field *Dialog control* is provided with the following configurations:

- Machine/workplace configuration
- Order type configuration
- Terminal configuration (not yet available via GUI)

In field *Dialog control*, you enter an "identifier", which controls the further processing.

These identifiers are used in the application *Dynamic dialogs - fields* in tab *Options* in the fields -Visible J/N

- Visible with dialog control/Not visible with dialog control

to integrate the required control of the input fields.

Functionality

An input field/control in a dynamic dialog is displayed, if it is

- visible = J

or

- visible = N, but the *Visible with dialog control* field includes a (sub) string transferred by the *Dialog control* field of one of the configurations. Several "dialog control identifiers" can be specified, separated by semicolon.

An input field/control is not displayed, if it is

- visible = N

or

- visible = J, but the *Not visible with dialog control* field includes a (sub) string transferred by the *Dialog control* field from one of the three configurations. Several "dialog control identifiers" can be specified, separated by semicolon.

Example:

- Machine 4711, Field *Dialog control* = M1
- Order type 0, Field *Dialog control* = A1

In the application *Dynamic dialogs - fields*, the configuration is as follows:

- [] Visible
- Visible if [A1;M]

The input field or control is displayed if either M1 or A1 or both is true. This means: The posting is made for a machine where the field *Dialog control* includes M1 and/or an order/OP is logged on where the order type is configured with A1 in field *Dialog control*.



The fields that are hidden via dialog control are not sent to the server via dialog strings.

Field attribute <FIELDPAGER>

With a FIELDPAGER, the AIP generates several input fields for a configured field in the entry dialog depending on a list on the AIP. This can be used to collect scrap with several scrap reasons for partial confirmations and to delete and interrupt operations. The FIELDPAGER then generates automatically an input field for a valid scrap reason on any machine.

The multiple entry is only suitable if you need not display more than 80 elements.

The following fields are used from the dialog configuration:

Identifier

The identification gets the prefix and no „\$CT.“ and no index. Example:

Scrap: „\$CT.AUS:“

Rework: „\$CT.NCH:“

Open quantity: „\$CT.PRB:“

If the machine status is entered hierarchically, the identification is MST.

Positions

Position Field	FPOSX and FPOSY	= Top left corner of the input object
Position Unit	EPOSX and EPOSY	= Width and height of the input object

Alignment

"Left"

Category

„Grid“

Length

Input length of multiple fields, e.g. 8

Create an input format

The input format by default is integer. If inputs with decimal places are required, the input format can be the same as for normal input fields with the input type "FLIESS" and the formatting "#####.##" (7 digits with 2 decimal places) or via a configured input type. The input types are restricted to numeric field types (with or without decimal places).

Field attribute 1

„FIELDPAGER“.

Dialog list function

Preconfigured fieldpager from a PAGER - section in the file ctaipplay.ini, for example "PAGER-AGRD.LST" (see below)

User defined 1

„AUTO“

The fields are generated using the font/size configurations designed for workflow fields (see "Dialog.ini")

- ➔ LabelFont... (Name, Size, Style, Color) (for multiple input field identifiers)
- ➔ FieldFont ... (Name, Size, Style, Color) (for multiple input field)

You can configure the available FIELDPAGER in a PAGER-section of the file ctaipplay.ini. The following configuration are prepared in the file ctaipplay.ini in the standard.

Section	Purpose
PAGER-AGRD.LST	Collection of scrap with several scrap reasons in case of partial confirmations, interruption and termination of operations. The FIELDPAGER generates a valid scrap reason for the input field at the machine.
PAGER-AGRD-NCH.LST	Collection of rework quantities with several reasons for partial confirmations or interrupting and terminating operations. The FIELDPAGER generates a valid rework reason for the input field at the machine.
PAGER-AGRD-PRB.LST	Collection of open quantities with several scrap reasons for partial confirmations or interruption and termination of operations. The FIELDPAGER generates a valid reason for open quantities for the input field at the machine.
MST-BUTTON-PAGER	Hierarchically input of the machine status.

Example for "Multiple scrap input".

Entry	Comment
Section [PAGER-AGRD.LST]	Configuration in workflows (dynamic dialogs) ➔ Field attribute:1 = FIELDPAGER ➔ List = PAGER-AGRD.LST
FILE=agrd.lst	Name of file used to generate multiple input fields
INI=	INI/configuration file including grid layout definition (Default = ctaipplay.ini)
SECTION=WF-PAGERPANEL-AGRD.LST	Section that includes the definition of the grid layout. The section "WF-PAGERPANEL-AGRD.LST" is required because the sorting does not work properly with ORDER sorting if file contents are unsorted. (Numeric sorting 1,2,3,4,5,10,11,22,... / alphanumeric 1,10,11,2.22.3.4.5....)

Entry	Comment
FILTER=MNR=<MNR> & ART=A	Possible filter to generate multiple input fields from the file <FILE>
ORDER=GRTXT	Possible sorting to generate multiple input fields from the file <FILE>
LabelColumn=GRTXT	Configuration of the column from the file <FILE> that has been designed as identifier for multiple input fields. (Default GRTXT)
LABELBEFOREFIELD=true	Optional: Identifier input field (Default=False)
LABELHEIGHTFAKTOR=1.25	Optional: Factor for calculating the height of a multi input field for positioning (default = 1.25)
LABELCHARCOUNT=20	Optional: Number of characters displayed for the label text of a multiple input field (see <i>LabelColumn</i>) (default=20)
IDCOLUMN=GR	Optional: Configuration of the column from the file <FILE> that includes the "KeyValue" of the row. (Default GR)
MODE=... SHOWIDCOLUMN=1 SHOWKEYLABEL=1 BUTTON-PAGER=1 or BUTTON-RESULT=1 BUTTON-RESULT=0 BUTTON-SKIN=BUTTON_BIG 	Extended options - shows (<IDCOLUMN>) in label (Default=0) - shows hotkeys 'a' .. 'z'(default=0) - item is shown as button (default=0) - button closes the dialog (default=1/active) ..as of V# 2.0.3.45 - the dialog remains open. The user exit „DynDlgFunctions_<dlg>“ with the function "ON(<KENN>)CLICK" is executed. - skin for button (default=BUTTON_BIG)
INCREMENT-BUTTON-COLOR=clSilver MODE=.. INCREMENT-BUTTON=TRUE	As of CTAIP V# 2.0.3.10 If the mode "INCREMENT-BUTTON=TRUE" is set, you can increment the value by 1 if you click on the label of the input field. Use the configuration „INCREMENT-BUTTON-COLOR=clSilver“ to change the label background. Default is "clSilver".

Entry	Comment
Section [WF-PAGERPANEL-AGRD.LST]	Configuration of a grid layout for the generation of multiple input fields
GR=N10,100,R	Definition that enables numeric sorting.
GRTXT=C10,200,L	Standard configuration for alphanumeric sorting.



If you use user exits at the AIP as part of the MES Development Suite, bear in mind that the DynDlgFieldchange event is not normally triggered for the fieldpager when a field is changed. This is required to calculate a sum field that might exist.

If the field attribute "**FIELDCHANGE**" is added to the Fieldpager, the event is also triggered for its input fields. You must insert the following line in DynDlgFieldchange_XYZ in your user exits to ensure that totals are still calculated.

Example for Fieldpager with ID \$CT.AUS::

```
Sub DynDlgFieldChange_A_TR
  Select Case VDlg ("DLG.FLD")
    Case "$CT.AUS:"
      DLGVAR=Item("$CT.AUS:SUM",VDlg("$CT.AUS:SUM"))
  End Select
End Sub
```

Field attribute <COLORLABELFONT>

You can use the field attribute <COLORLABELFONT> to display texts with a special font. (see "Dialog.ini" -> ColorLabelFont[Name,Size,..])

The default field color is < blue > and can be used for the following dynamic dialog fields

GRID	- for Grid/Image/Shape/... Header
IMAGE	
SHAPE	
FIELDPAGER	

TEXTVIEW

METER

TEXT	- for text display
Input	- for field label

Use {c~color} to switch the color.

For example: "Log on {c~clblack} <ANR> {c~clblue} order"
 → "Log on AU0010001AG01 order"

The color configuration may also be used for the workflow caption. Here, the default font color is black.



Translation: If you want to translate the new texts/labels, you must add the texts to the relevant translation file and translate them (standard: ctaip.mld, custom: ctaipkd.mld).